

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Slug Catcher Preliminary Specification
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-SPC-001
Document Revision	R01
Document Status	Issued for Review
Originator	Ridwan Olagoke
Security Classification	Restricted
Issue Date	7-Jul-22

TATA CONSULTING ENGINEERS LIMITED	
CONTRACTOR DOCUMENT REVIEW STATUS	
☐	1. Approved, further work can Proceed
☐	2. Approved with comments. Work can proceed subject to incorporation of comments
☐	3. Rejected. Revise according to comments & resubmit
☒	4. Retained for Information
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Rev	SMY-05.08.2022

Slug Catcher Preliminary Specification

R01	07.07.22	Issued for Review	O. Ridwan	K.Yusuff	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	07-07-22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-SPC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

The document, in consideration of relational design elements & concepts, forms the technical basis for the development of the facilities Project Specifications & tendering of contracts for the project's implementation.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
	FISAS Draft ESIA Report (November 2020)

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)
2.0	PMC	TATA CONSULTING ENGINEERS LIMITED (TCE)
3.0	EPCC (CONTRACTOR)	CHINA TIANCHEN ENGINEERING CORPORATION (TCC)
4.0	Project Title	Gas Processing and Methanol Complex
5.0	Location	Akabel, Brass Island. The project site is about 85km South West of Port Harcourt, 70km South of Yenagoa and 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port and distance	Port Harcourt Airport (PHC) Bayelsa Airport (Under Construction)
5.3	Nearest Port	Port Harcourt (Onne)
5.4	Nearest Railway Station & distance	-
5.5	Highway and distance	Connected through water way only. Road connection does not exist.
5.6	Important Town and distance	Yenagoa, Nigeria & 67 km
5.7	Altitude Above Mean Sea Level	9 m
5.8	Atmospheric pressure	
	Average	1013 mbar
6.0	Ambient Air Temperatures	(Refer Note-5,6)
6.1	Maximum	38°C
6.2	Minimum	18°C
6.3	Average	23°C
6.4	Summer Design Wet Bulb Temp	28°C
6.5	Winter Design Wet Bulb Temp	15°C
6.6	MDMT	15°C

Project Information – General (Contd)

6.7	Swamp temperature	250C. (230C. to be used wherever a margin is required)
6.8	Ground temperature	25 - 27.50C.
6.9	Black-bulb temperature:	800C.
7.0	Relative Humidity	
7.1	Maximum	95 %
7.2	Minimum	51 %
7.3	Average	73 %
8.0	Rainfall Data	
8.1	Average annual rainfall	3800mm
8.2	Max daily rainfall:	180mm
8.3	Mean max. Hourly rainfall	100mm
9.0	Transport	
9.1	Name of the highway near which the plant is located	Connected through water way only. Road connection does not exist.
9.2	Access road	-
9.3	Railway (Gauge)	-
10.0	Wind Data	
10.1	Design Wind Velocity	35.6 m/sec (Note-1)
10.2	Wind Direction	Note-2
11.0	Seismic Data	
11.1	Earthquake Zone	Note-3
12.0	High Flood Level (HFL)	*
13.0	Natural Ground Level	*
14.0	Slope	*
15.0	Drainage	*
16.0	Design temperature for Electrical equipment	400C

Definition of Terms

TERMS	DEFINITIONS
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase I)

Abbreviations

ABBREVIATION	MEANING
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity

Abbreviations (contd.)

ABBREVIATION	MEANING
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System

Abbreviations (contd.)

ABBREVIATION	MEANING
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Triethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

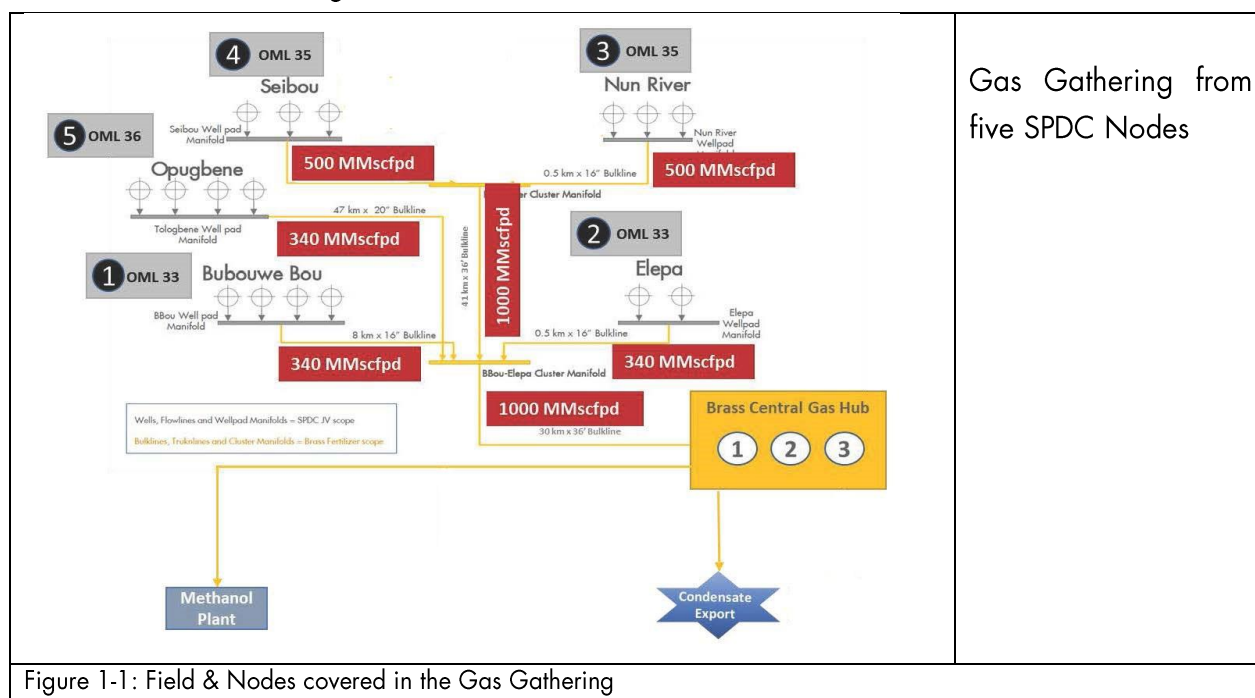
Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

The Nodal Gas Gathering Schematics



The Nodal Gas Gathering Schematics (contd.)

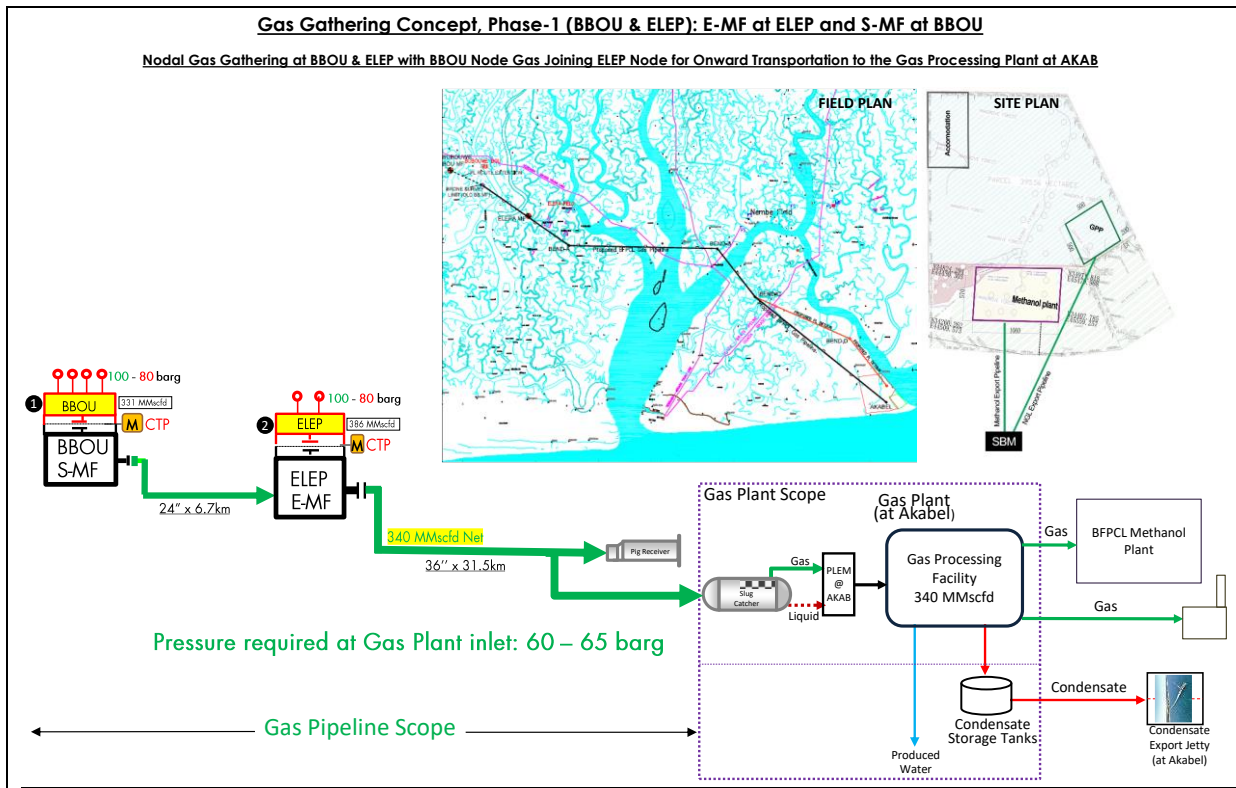


Figure 1-2: Gas Gathering Concept, Phase 1

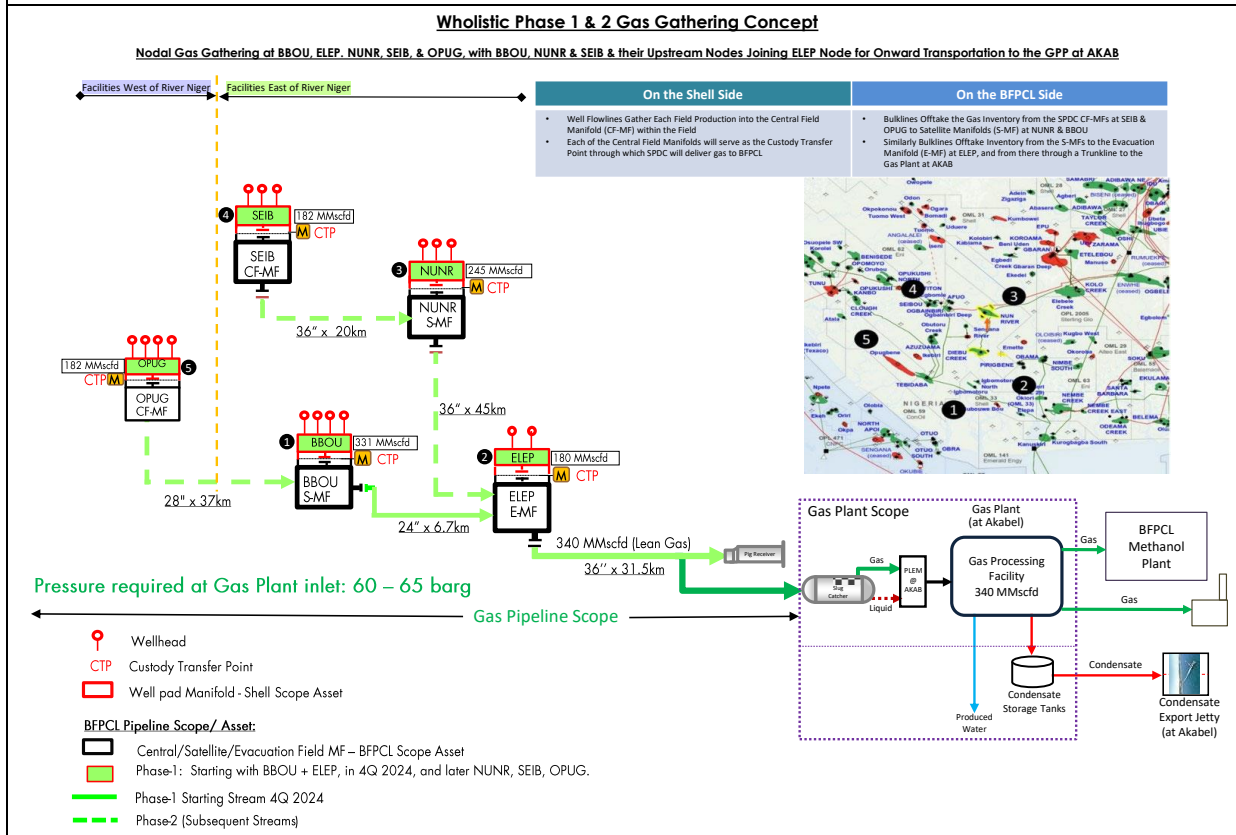


Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 & 2

1.2. Purpose of Document

The purpose of this document is to provide the preliminary design, selection criteria and specification for the slug catcher for the development of the Gas gathering Project.

The gas gathering will be in two phases, as shown in the schematics below:

- a. Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
- b. Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

1.2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and
- The Pipeline End Manifold at AKAB.

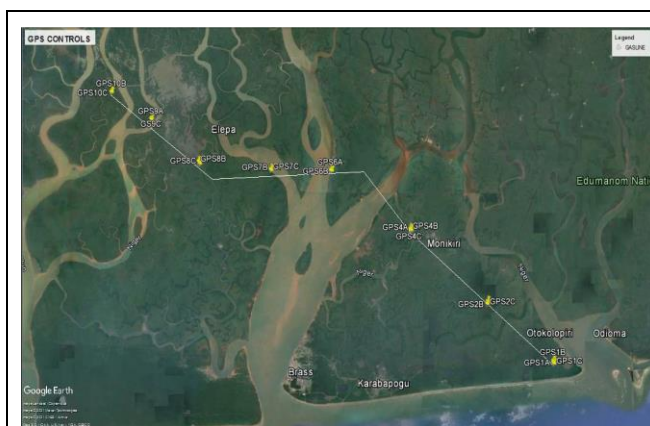
These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED. The phase 2 pipelines routes are yet to be defined and surveyed.

1.2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

1.3. The Phase 1 Pipelines Route / Right of Way



The pipeline route / Right of Way, running from BBOU through ELEP to AKAB is shown in the satellite image below

Figure 1-3: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

1.4. Scope

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

1.5. Design Battery limits

The battery limits for this Phase 1 FEED Project are:

1.5.1.BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

1.5.2.ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

1.5.3.Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

2. Regulations, Codes And Standards

2.1. Regulations

The following listed regulations in the below table will be used in the project as applicable.

Regulation	Function
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is BFPCL statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.

2.2. References

The following listed reference documents in the below table will be used in the project as applicable.

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-	Manifold Platforms Concept & Integration with SPDC

EA/SPDC-BFPC UGS-GEP/PRC-RPT-	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.
DEP 31.40.40.38-Gen.	Hydrostatic pressure testing of new pipelines
DEP 33.64.10.ok g10-Gen.	Electrical engineering guidelines
DEP 61.40.20.30-Gen	Welding of pipelines and related facilities

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Slug Catcher Preliminary Specification

3.1. Type Selection

Pipe-Finger Slug Catcher is selected as most suitable type, following Type Selection Decision Tree developed by BP, and widely used in the Oil & Gas Industry. The red line in the tree shows the flow of decision leading to the selection of the finger type slug catcher for this Project.

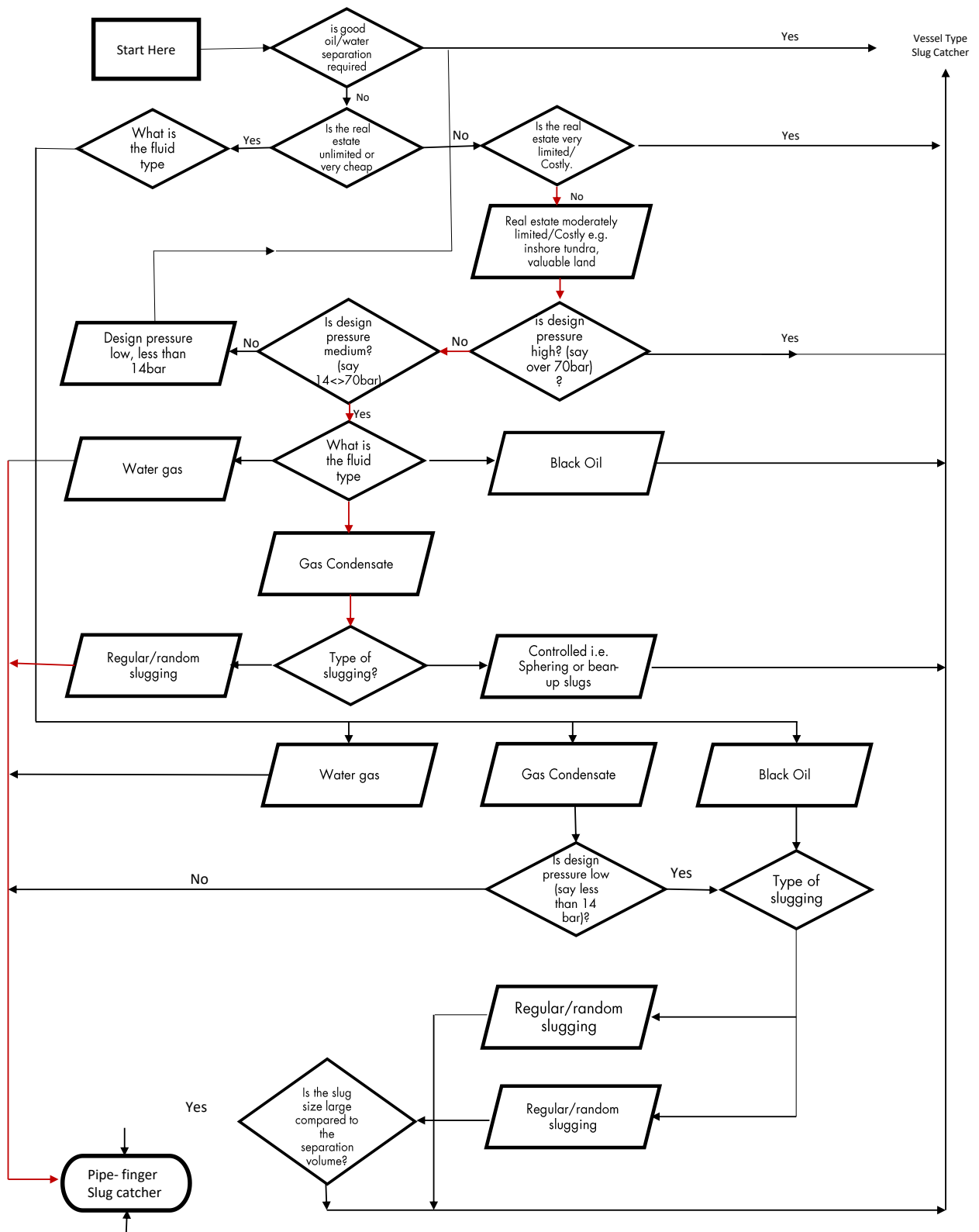
3.1.1. Considerations

- a. Large Volume of Liquid will be encountered ($\pm 5,782 \text{ m}^3$)
- b. Initial Separation for the Large Volume of Liquid to ensure efficient operation of inlet separators

3.1.2. Type Selection

- a. Two-Phase Separation – coarse separation
- b. Multi-pipe selected as most appropriate following Type Selection Tree (BP Selection Process)
 - i. Fluid Type (water gas rather than black oil)
 - ii. Large Slug Volume of $\pm 5,782 \text{ m}^3$ - ($> 100 \text{ m}^3$ ref Shell DEP)
 - iii. Moderate Pressure of 60 - 65 barg, - (14 barg $< P < 70$ barg, ref BP Type Selection Decision Tree)
 - iv. Cost Advantage over high pressure vessel design (view smaller wall thickness for pipe compared with vessel for a given pressure)
 - v. Greater flexibility in design possibilities to cope with wide ranges of flow situations
 - vi. Sizing: - Sized for max Slug Volume

BFPF SLUG CATCHER TYPE SELECTION DECISION TREE CHART



Culled from – BP Multiphase Manual

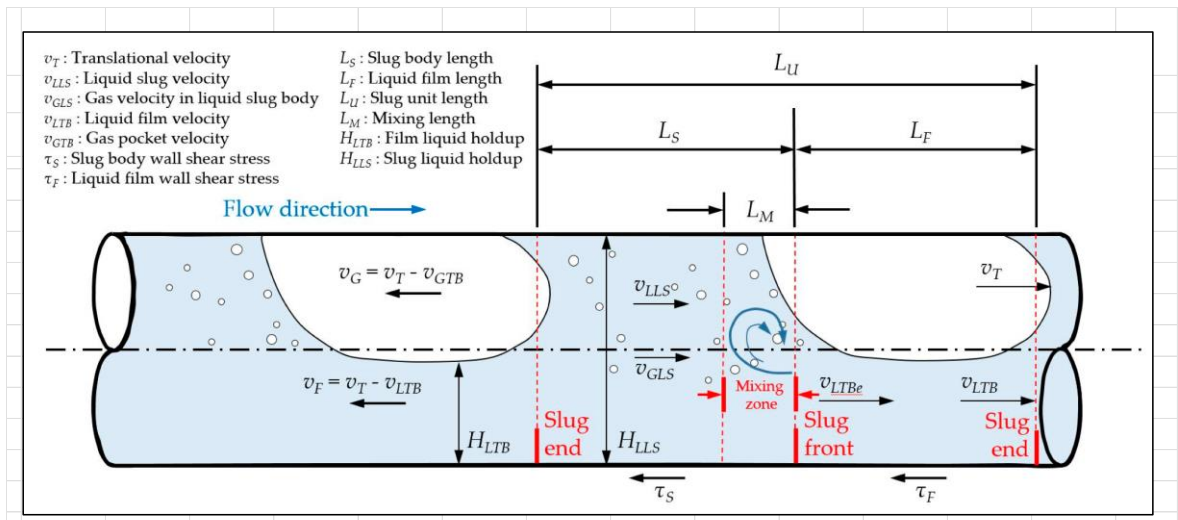
Figure 5.1: Slug Catcher Type-Selection Decision Tree

3.2. Slug Volume & Slug Catcher Calculations

The calculation for the slug volume and slug catcher was based on the maximum flow of gas at a pressure of 100 barg. Slug generated from BBOU-ELEP and ELEP-AKAB pipelines were cumulated to form the expected slug volume. A margin of error of 20% was added to account for process upsets and other unforeseen incidents that could increase the slug volume.

3.2.1. Liquid Hold-up Calculations

LIQUID HOLDUP VOLUME CALCULATION			
100 barg Pressure			
Parameters	BBOU-ELEP Pipeline	ELEP-AKAB Pipeline	Units
Pipeline Outside Diameter	610	914	mm
	578.24	866.34	
Pipeline Internal Diameter	0.57824	0.86634	m
Pipeline Area	0.263	0.589476661	m ²
Mean Liquid Hold Up	0.1644	0.261280054	
Pipeline Length	6662.74	35676.75	m
Liquid holdup Volume	287.5960	5494.879293	m ³
Expected Liquid Hold Up Volume		5782.4753	m³
Parameters		2027 - 340MMSCFD	Units
Expected Slug Volume (20% margin)		245047.427	#3



3.2.2. Calculations Notes

3.2.2.1. Fluid Properties

3.2.2.1.1. Properties From Simulation Program

The following properties of the fluid were determined from the HYSYS Simulation for maximum gas flow of 340 MMscfd.

3.2.2.1.2. Table 1: Fluid Properties from HYSYS Simulation Derived Properties

Using information from Table 1, following relevant properties of the fluid needed for the slug catcher sizing were derived using formula stated.

1. Total Liquid Mass Flow Rate: $W_L = W_C + W_W$

2. Density of Liquid: $\frac{W_L}{\frac{W_C}{\rho_C} + \frac{W_W}{\rho_W}}$

3. Volumetric Flow Rate of Gas: $\frac{W_g}{3600 \times \rho_g}$

4. Volumetric Flow Rate of Liquid:

$$\frac{W_L}{3600 \times \rho_L}$$

3.2.2.1.3. Summary of Derived Properties

Property	Sym- bol	Unit	Calcu- lated Value

Total Liquid Mass Flow Rate	W_L	lb/hr	651410.37
Density of Liquid	ρ_L	lb/ft^3	42.46
Volumetric Flow Rate of Gas	Q_g	ft^3/s	45.10
Volumetric Flow Rate of Liquid	W_l	ft^3/s	4.261

3.2.2.2. Separation Part Sizing

1. $C'Re^2$

$$C' (Re)^2 = \frac{(0.95) (10^8) \rho_g D_p^3 (\rho_l - \rho_g)}{\mu^2}$$

2. Drag Coefficient

The drag coefficient was determined from the following chart, culled from GPSA Handbook, Chapter 7.

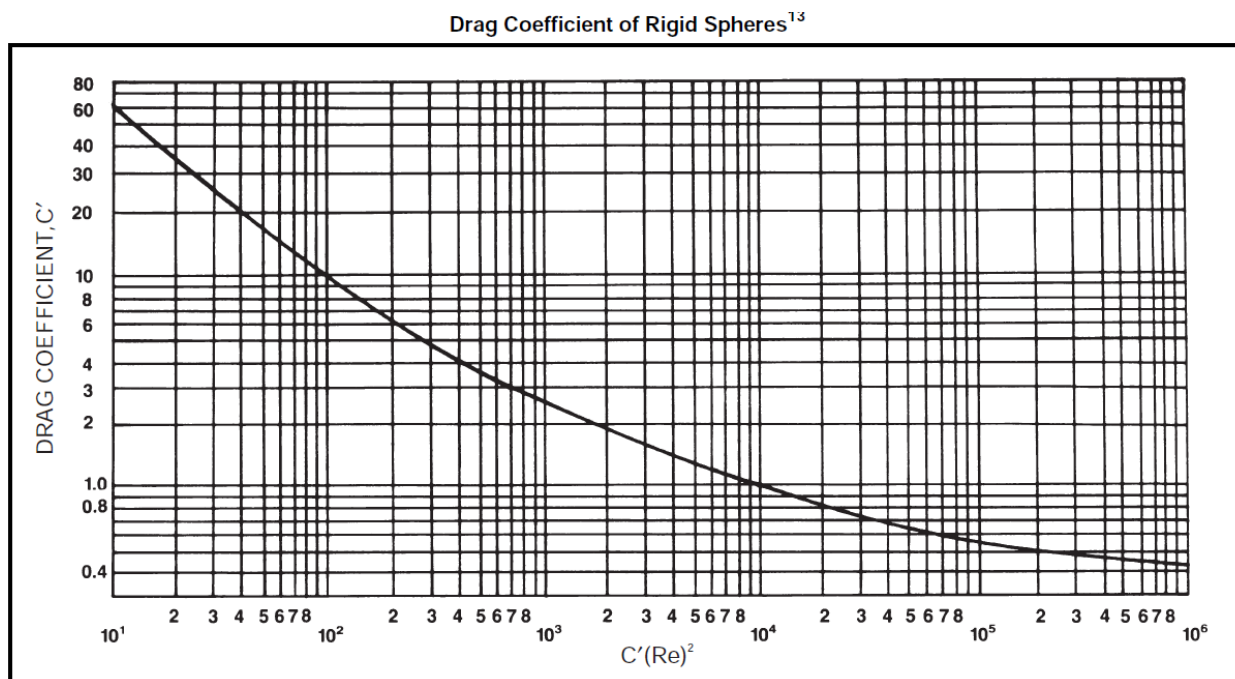


Figure 5.2: Reynold's No. of Particle

$$Re_p = \frac{C' Re_p^2}{C'}$$

3. Terminal Velocity

$$V_t = \sqrt{\frac{2 g M_p (\rho_l - \rho_g)}{\rho_l \rho_g A_p C'}} = \sqrt{\frac{4 g D_p (\rho_l - \rho_g)}{3 \rho_g C'}}$$

4. Proportion of Bottle for Liquid Storage and Gas Handling

It was assumed that 70% of the cross-sectional area of the primary bottle will be used for liquid storage while the remaining 30% will be used for gas handling.

5. Selected Primary Bottle Size

The nominal size of the primary bottles was selected based on acceptable industry practice of 1.5 x Pipeline size. The wall thickness is same as the pipeline wall thickness.

6. Cross Sectional Area Available for Gas Flow

$$(\text{Proportion of Bottle for Gas Handling}) \times (\text{Bottle Cross sectional Area})$$

7. Vertical Height of Bottle Occupied by Gas

Vertical height of bottle occupied by gas is found by iteration using the following formular.

$$\text{Area occupied by Gas, } A_v = \left(\frac{D_b}{2}\right)^2 \cos^{-1}\left(\frac{\left(\frac{D_b}{2} - d_g\right)}{\left(\frac{D_b}{2}\right)}\right) - \left(\frac{D_b}{2} - d_g\right) \sqrt{2\left(\frac{D_b}{2}\right)d_g - d_g^2}$$

8. Settling Time of Liquid Droplets

$$\text{Settling time, } t = \frac{d_g}{V_t}$$

9. Limiting Gas Velocity inside Bottle

Industry best practice has it that settling of droplets can occur by gravitation in a primary bottle when flowing at low velocities, typically at less than 2 m/s, towards the gas riser, provided sufficient residence time is allowed for it to happen. For this design, 1 m/s was adopted.

10. Number of Primary Bottles

$$N = \frac{4(Q_L + Q_G)}{\pi D_b^2 V_g}$$

The number of primary bottles, Nb is rounded up to the nearest power of 2.

$$L_e = \frac{4 \times t \times Q_g}{\pi D_b^2}$$

11. Selected Number of Splitters

For efficient filling and maximum utilization of storage capacity of the slug catcher, the fluid flow was distributed equally between parallel bottles using splitters in the form of Tee-junctions with the

flow impacting perpendicularly at the tee-junction before being divided into equal streams departing through the runs. The number of splitters is selected to limit the number of parallel fingers in each split to 4.

Summary of Separation Part Sizing Calculation

Property	Sym- bol	Unit	Calcu- lated Value
C'Re²			8.98E+03
Drag Coefficient	C'		1.12
Reynold's Number of Particle	Re _p		8.02E+03
Terminal Velocity	V _t	ft/s	0.400
Proportion of Bottle for Liquid Storage			70%
Proportion of Bottle for Gas Handling			30%
Selected Primary Bottle			
Outside Diameter	OD	in	48
Wall Thickness	WT	in	0.94
Inside Diameter	D _b	ft	3.843
Bottle Cross sectional Area	A _b	ft ²	11.601
Cross Sectional Area Available for Gas Flow	A _v	ft ²	3.480
Vertical Height of Bottle Occupied by Gas	d _g	ft	0.717
Settling Time	t	s	1.793
Limiting Gas Velocity inside Bottle	V _g	ft/s	1
Number of Primary Bottles Required	N		4.255
Selected Number of Primary Bottle	N _b		24
Length of Pipe Required for Separation	Le	ft	6.97
Selected Number of Splitter	N _s		6

3.2.2.3. Intermediate Section Length Sizing

The intermediate section is located between the gas-liquid separation section and the slug-receiving section. It was designed based on the prohibition of liquid entry to the gas riser.

Industry best practices allow for selection of slope of this section to be between 2.5% and 5%. For this design, 2.5% was selected.

1. Length of Intermediate Section

$$L_s = \left(\frac{D_b}{\tan \theta} \right)$$

Where, θ is the angle of slope of section with riser.

Intermediate Section Part Sizing			
Angle of Slope of Intermediate Section	θ		2.50%
Horizontal Length of Intermediate Section	Ls	ft	153.733333

2. Slug Receiving Section Length

1. Expected Slug Volume Calculation

Liquid hold-up volume was calculated from:

(Mean Liquid Hold Up over entire pipeline) x (Pipeline Area) x (Pipeline Length)

Expected Liquid Hold-up was calculated as the sum of liquid hold-up volumes in each pipeline segment (BBOU-ELEP and ELEP-AKAB Pipelines)

Expected slug volume was then calculated as 120% of Expected Liquid Hold-up to cater for liquid carryout during slug arrival.

	100 barg Pressure		
Parameters	BBOU-ELEP Pipeline	ELEP-AKAB Pipeline	Units
Pipeline Outside Diameter	610	914	mm
	578.24	866.34	
Pipeline Internal Diameter	0.57824	0.86634	m
Pipeline Area	0.263	0.589476661	m ²
Mean Liquid Hold Up	0.1644	0.261280054	
Pipeline Length	6662.74	35676.75	m
Liquid holdup Volume	287.5960	5494.879293	m ³
Expected Liquid Hold Up Volume		5782.4753	m³
Parameters	2027 - 340MMSCFD		Units
	Expected Slug Volume (20% margin)	245047.427	ft³

2. Bottle Slope in Slug Receiving Section

Industry best practice allows for selection of slope of this section to be between 1% and 1.5%. For this design, 1.2% was selected.

3. Partially Filled Volume of Slug Catcher

For the purpose of determining the maximum capacity of the slug catcher, the volume in the partly filled part of the bottle can be assumed to be:

$$V_{part} = \frac{\pi (D_b^3)}{8 \tan \alpha}$$

4. Residence Time

The time over which liquid surges have to be coped with is determined by the nature and operating philosophy of the system. It is a prerogative of the client and/or designer. The residence time is selected as 10 mins for this design.

5. Buffer Volume

Flow rate of Liquid x Residence Time

6. Total Anticipated Holding Volume

$$V_{tot} = V_{slug} + V_{buffer} - \text{Partially Filled Volume}$$

7. Length of Slug Receiving Section

$$L_t = \frac{4.8 \times (V_{tot})}{\pi D_b^2 N_b}$$

8. Summary of Slug Receiving Section Length Sizing

Slug Receiving Section Length Sizing			
Property	Symbol	Unit	Calculated Value
Expected Slug Volume	V_{slug}	ft ³	206441.489
Partially Filled Angle	α		1.20%
Partially Filled Volume	V_{part}	ft ³	1857.81642
Assumed Residence Time	t_{res}	s	600
Buffer Volume	V_{buffer}	ft ³	2556.85232
Total Holding Volume	V_{tot}	ft ³	207140.525
Length of Slug Receiving Section	L_t	ft	855.55057
Slug Catcher Total Length	L_{total}	ft	1016.2541

3.2.2.4. Gas Riser Sizing

1. Load Factor Parameter

The separation capability of a riser is determined by the load factor. Shell DEP specifies that load factor must be less than 0.20 m/s (0.6 ft/s) so that droplets larger than 2 mm will settle out from the stream. For this design, load factor of 0.4 ft/s was considered.

2. Superficial Velocity of Gas

$$V_{SG} = \xi \sqrt{\frac{\rho_L - \rho_g}{\rho_g}}$$

3. Required Inside Diameter of Riser

$$D_{riser} = \sqrt{\frac{Q_g}{N_{rb} \left(\frac{\pi}{4}\right) N_b V_{SG}}}$$

4. Selected Riser Diameter

A pipe that has internal diameter greater than calculated and having minimum wall thickness to maintain mechanical integrity was selected.

5. Height of the Riser above Separation Bottle

The Shell DEP states that “the riser should have a minimum height to allow liquid entrained in the riser gas stream to settle. The riser height should be at least 5 times its diameter or 5 m, whichever is less”.

6. Summary of Gas Riser Sizing Calculation

Property	Sym- bol	Unit	Calcu- lated Value
Load Factor Parameter	ξ		0.4
Superficial Velocity of Gas	Vsg	ft/s	1.16408803
Number of Riser Per Bottle	Nrb		1
Required Inside Diameter of Riser	Driser	ft	1.43361249
Selected Riser Diameter	Riser		NPS 36
Height of Riser	Hriser	ft	7.16806243

3.2.3. Slug Catcher Calculations Summary Results

Fluid Properties (Derived from HYSYS)			
Property	Symbol	Unit	Calculated Value
Gas Mass Flow Rate	W_g	lb/hr	727996.33
Condensate Mass Flow rate	W_c	lb/hr	649974.04
Water Mass Flow rate	W_w	lb/hr	1436.33
Density of Gas	ρ_g	lb/ft ³	4.48

Density of Condensate	ρ_c	lb/ft ³	42.43
Density of Water	ρ_w	lb/ft ³	61.98
Acceleration Due to Gravity	g	ft/s ²	32.2
Particle Diameter	D_p	micron	150
Viscosity of Gas	μ	cP	1.47E-02

Fluid Properties (Calculated)

Property	Symbol	Unit	Calculated Value
Total Liquid Mass Flow Rate	W_L	lb/hr	651410.37
Density of Liquid	ρ_L	lb/ft ³	42.46
Volumetric Flow Rate of Gas	Q_g	ft ³ /s	45.10
Volumetric Flow Rate of Liquid	W_l	ft ³ /s	4.261

Separation Part Sizing

Property	Symbol	Unit	Calculated Value
$C'Re^2$			8.98E+03
Drag Coefficient	C'		1.12
Reynold's Number of Particle	Re_p		8.02E+03
Terminal Velocity	V_t	ft/s	0.400
Proportion of Bottle for Liquid Storage			70%
Proportion of Bottle for Gas Handling			30%
Selected Primary Bottle			
Outside Diameter	OD	in	48
Wall Thickness	WT	in	0.94
Inside Diameter	D_b	ft	3.843
Bottle Cross sectional Area	A_b	ft ²	11.601
Cross Sectional Area Available for Gas Flow	A_v	ft ²	3.480
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Limiting Gas Velocity inside Bottle	V_g	ft/s	1
Number of Primary Bottles Required	N		4.255
Selected Number of Primary Bottle	N_b		24
Length of Pipe Required for Separation	L_e	ft	6.97
Selected Number of Splitter	N_s		6

Intermediate Section Part Sizing

Angle of Slope of Intermediate Section	θ		2.50%
Horizontal Length of Intermediate Section	L_s	ft	153.733333

Slug Receiving Section Length Sizing			
Property	Symbol	Unit	Calculated Value
Expected Slug Volume	V_{slug}	ft^3	206441.489
Partially Filled Angle	α		1.20%
Partially Filled Volume	V_{part}	ft^3	1857.81642
Assumed Residence Time	t_{res}	s	600
Buffer Volume	V_{buffer}	ft^3	2556.85232
Total Holding Volume	V_{tot}	ft^3	207140.525
Length of Slug Receiving Section	L_t	ft	855.55057
Slug Catcher Total Length	L_{total}	ft	1016.2541

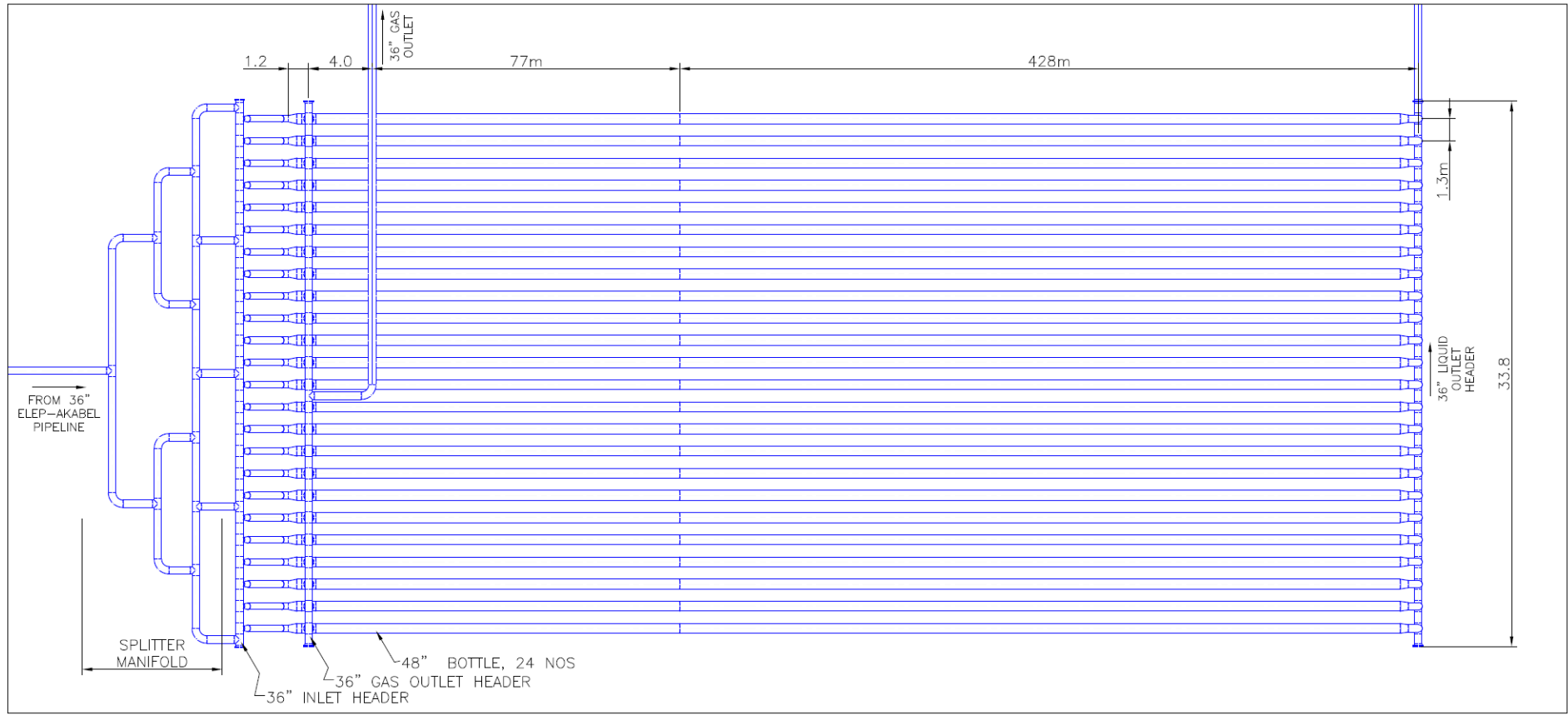
Gas Riser Sizing			
Property	Symbol	Unit	Calculated Value
Load Factor Parameter	ξ		0.4
Superficial Velocity of Gas	V_{sg}	ft/s	1.16408803
Number of Riser Per Bottle	N_{rb}		1
Required Inside Diameter of Riser	D_{riser}	ft	1.43361249
Selected Riser Diameter	Riser		NPS 36
Height of Riser	H_{riser}	ft	7.16806243

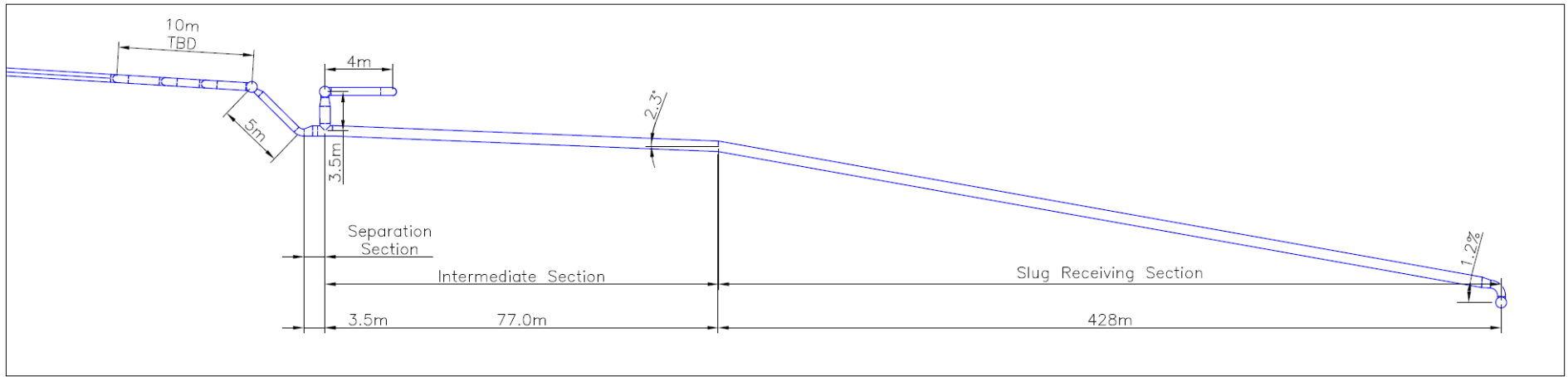
3.3. Slug Catcher Summary

S/N	Parameter	Value	Units
1	Total Length of Slug Catcher	1016.3	ft
	Length of Pipe Required for Separation	7.0	ft
	Horizontal Length of Intermediate Section	153.7	ft
	Length of Slug Receiving Section	855.6	ft
2	Primary Bottle		
	Outside Diameter	48.0	in
	Selected Number of Primary Bottle	24.0	
3	Gas Riser		
	Selected Riser Diameter	NPS 36	
	Height of Riser	7.2	ft
4	Total Capacity of Slug Catcher	36893.3	bbl
5	Estimated Width of Slug Catcher	138.0	ft

3.4. The Slug Catcher Sketch

Plan & Sectional Views of the Slug Catcher is shown below





4. Mechanical Design

4.1. General

The Slug catcher is an integral part of the pipeline system and thus shall meet the requirements of DEP 31.40.00.10-Gen. for category D fluid. The demarcation of the slug catcher with downstream facilities shall be indicated on the process engineering flow schemes.

4.2. Maximum Allowable Operating Pressure

The MAOP of the slug catcher shall not be less than the MAOP of the feeding pipeline system if line packing is required for the operation of the pipeline system. If line packing is not required, then the MAOP of the slug catcher should still equal the MAOP of the feeding pipeline system as this will negate the requirement for separate overpressure protection.

Alternatively, if line packing is not required the MAOP of the slug catcher may be lower than for the feeding pipeline system if a significant net lifecycle cost benefit can be demonstrated from lowering the MAOP and installing dedicated overpressure protection for the slug catcher.

4.3. Design Factor For Hoop Stress

The design factor to limit the hoop stress arising from pressure containment shall be selected according to DEP 31.40.00.10-Gen.

4.4. Other Stress

All loads shall be considered when determining combined stresses during pressure testing and operations. These loads will include:

- pressure;
- thermal expansion;
- passage of slugs or surges;
- foundation and support reaction;
- settlement;
- environmental loads.

Deformations due to the possible different settlements of the slug catcher and the connecting structures shall be accounted for. For buried slug catchers, possible deformations predicted at the transitions between exposed and buried parts, due to different settlements of the slug catcher and the connecting structures, shall also be accounted for.

The total longitudinal stress due to the internal pressure and weight of catcher including content shall not exceed $2/3$ of SMYS. The equivalent stresses shall not exceed the limits specified in DEP 31.40.00.10-Gen.

4.5. Supports and Anchors

ANSI/ASME B31.8, shall apply for the design of supports and anchors. The foundation of locally supported slug catchers shall be in accordance with DEP 34.19.20.31-Gen. The bottles of a buried slug catcher should be situated entirely below original ground level in areas with a natural slope similar to the downward slope necessary for the gravity fill of the bottles.

On a level plot, the bottles may be buried in a stable sand mound above the original ground level, with the liquid outlet header above the original ground level, or to reduce the height of the mound, in a shallow pit.

4.6. Material Selection

4.6.1. Linepipe

Linepipe used for bottle sections of the slug catcher shall be in accordance with DEP 31.40.20.37-Gen.

4.6.2. Fittings

Fittings shall be in accordance with DEP 31.40.21.30-Gen.

4.6.3. Flanges

Flanges shall be specified on the basis of ASME B16.5 and/or MSS SP-44.

4.6.4. Valves

Valves shall be in accordance with API Spec 6D.

4.6.5. Prevention of brittle fracture

The selection of slug catcher materials exposed to low temperatures shall be in accordance with DEP 30.10.02.31-Gen.

A reference minimum metal temperature shall be established under the most adverse combination of depressurization and climate. The lowest temperature occurring in the slug catcher during depressurization shall be determined, taking into account the heat content of the slug catcher wall and the heat influx from outside.

Under certain circumstances the lower design temperature can be taken to be 50 °C above the reference minimum metal temperature. A detailed evaluation of the limiting values of primary local membrane stresses, accounting for the time-dependent relation between temperature and pressure during depressurization, shall be performed to establish the applicability of the 50 °C relaxation.

4.7. Welding

Welding shall be in accordance with DEP 61.40.20.30-Gen.

4.8. Internal/External Corrosion Protection

The potential for internal corrosion in the slug catcher shall be evaluated to determine the need for a corrosion allowance or other measures to mitigate internal corrosion.

Cathodic protection of a buried slug catcher shall be in accordance with DEP 30.10.73.31-Gen.

Anti-corrosion coating of both buried and non-buried slug catchers shall be in accordance with DEP 31.40.00.10-Gen.

4.9. Overpressure Protection

Separate dedicated overpressure protection shall not be required if the MAOP of the slug catcher is not less than the MAOP of the feeding pipeline system.

If the MAOP of the slug catcher is less than the MAOP of the feeding pipeline system then it shall be safeguarded against overpressure by full flow relief protection in accordance with DEP 31.40.10.14-Gen.

4.10. Maintenance

Ancillary equipment, such as PSVs, blowdown valves and liquid level shut down valves, which may require replacement shall be provided with the necessary isolation to permit maintenance without the shutdown and/or decommissioning of the slug catcher and other relevant facilities.

Both ends of headers shall be provided with end flanges for inspection access and for providing ventilation during the inspection. Both ends of liquid headers shall be provided with end flanges to facilitate the removal of sludge.

Requirements for the condition monitoring of buried slug catchers shall be established during the detail design stage and be designed to minimise the requirement for future excavation.